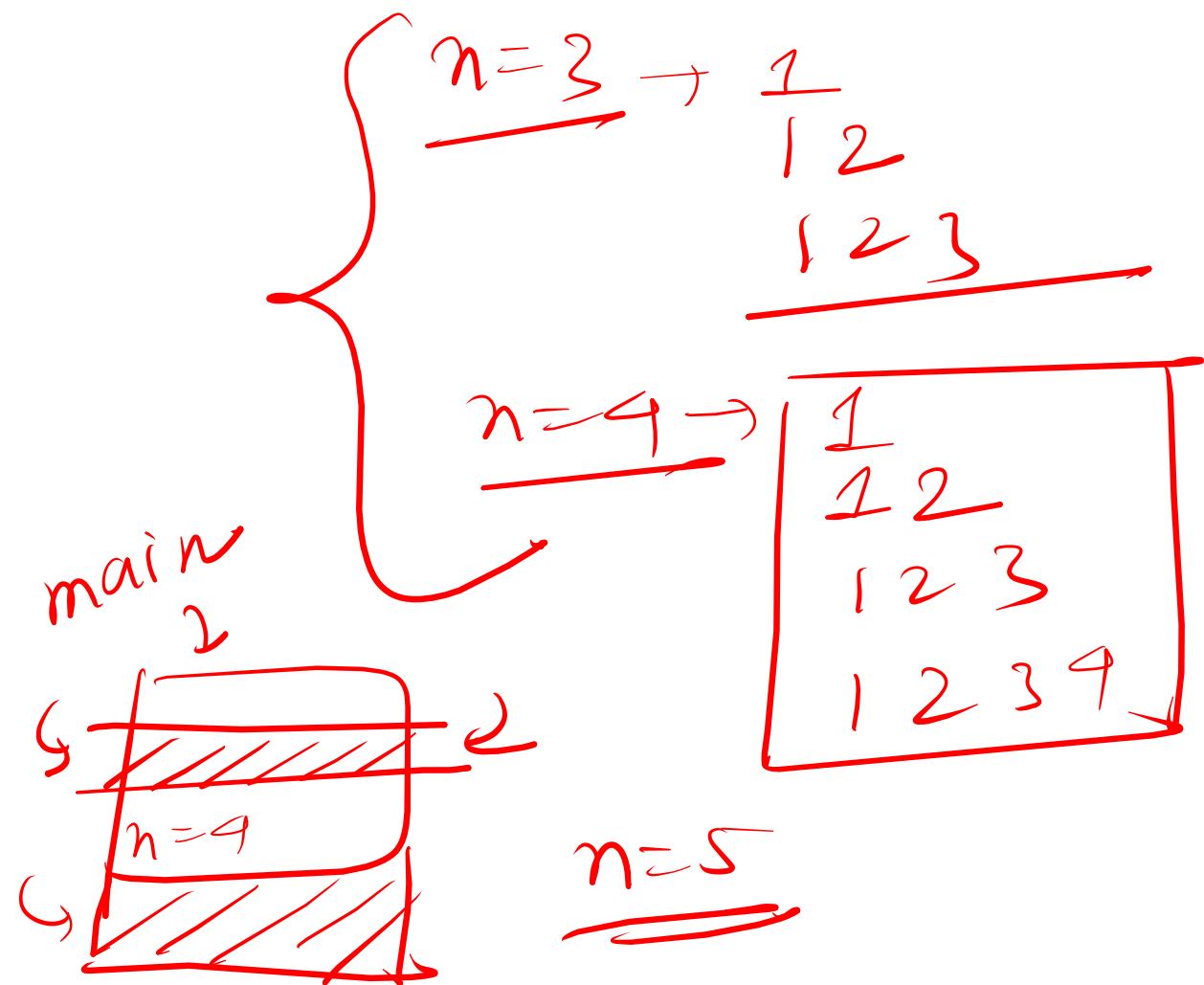


$$\begin{array}{l}
 x = \cancel{1} \cdot 2 \\
 \hookrightarrow (\cancel{1+1}) \rightarrow 2 \\
 \hline
 x + \cancel{1} \quad 2 \\
 \hline
 x = \cancel{2} \rightarrow 3
 \end{array}$$

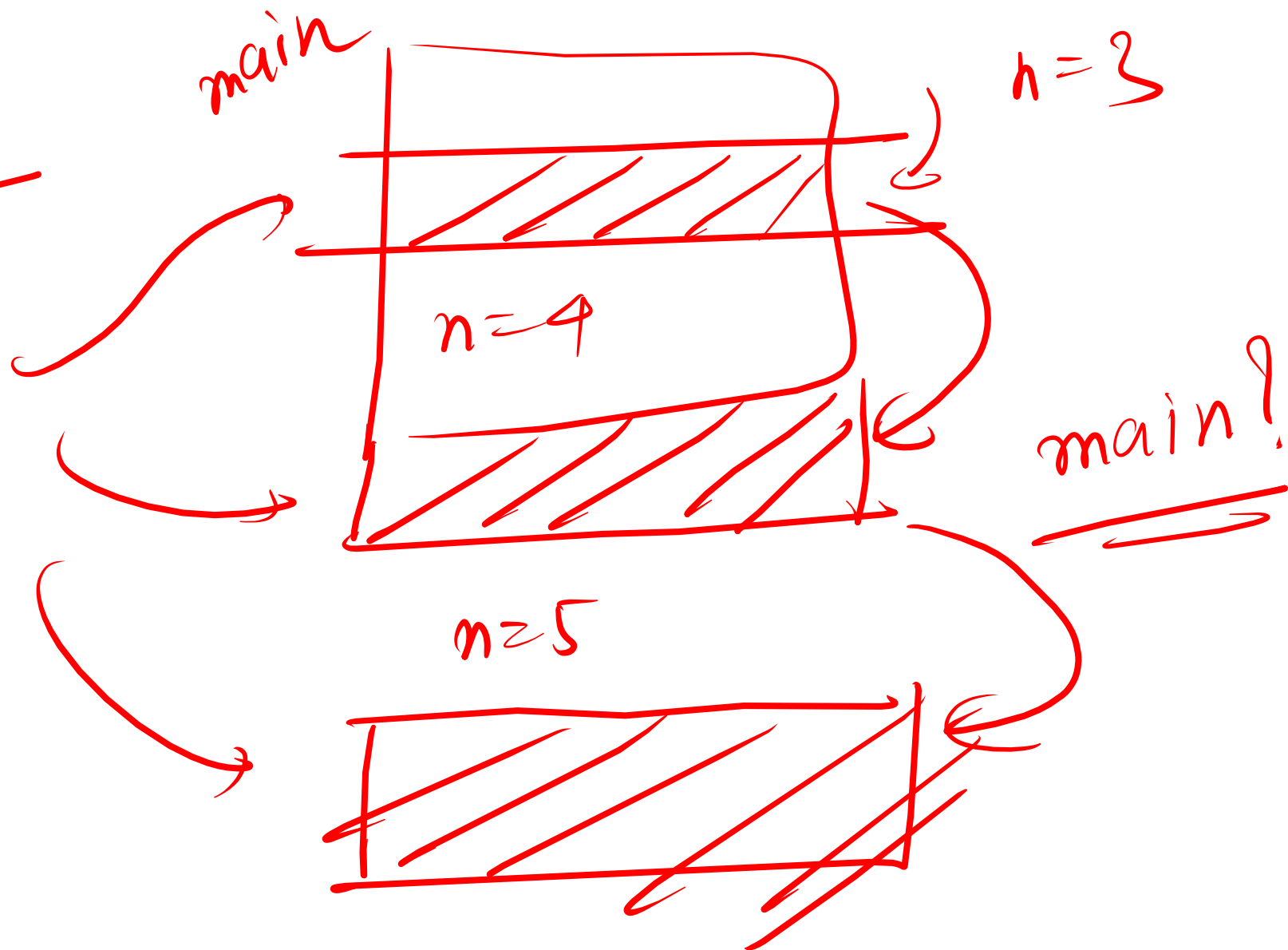
$$\begin{array}{l}
 \textcircled{+ - x -} \quad \textcircled{x} = \cancel{2} \textcircled{3} \\
 \textcircled{3} \\
 \hline
 x = \cancel{3} / \quad \cancel{x + 1} \\
 \quad \quad \quad 3
 \end{array}$$

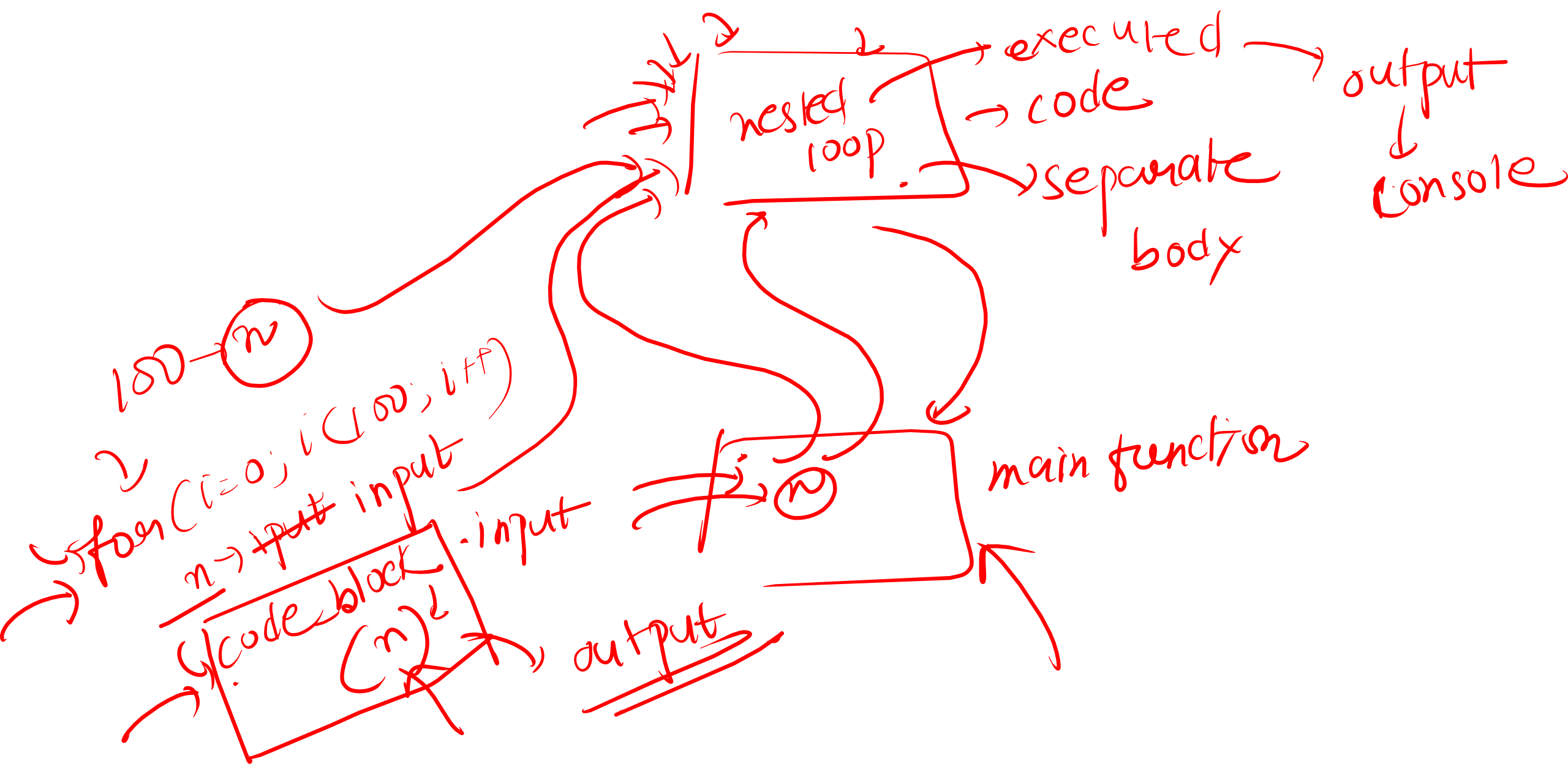
$n=3$
 function
 $\text{for}(i=1; i \leq n; i++) \{$
 $\text{for}(j=1; j \leq i; j++) \{$
 $\text{print}(j + " ");$
 $\}$
 $\text{print}("\n");$
 $\}$

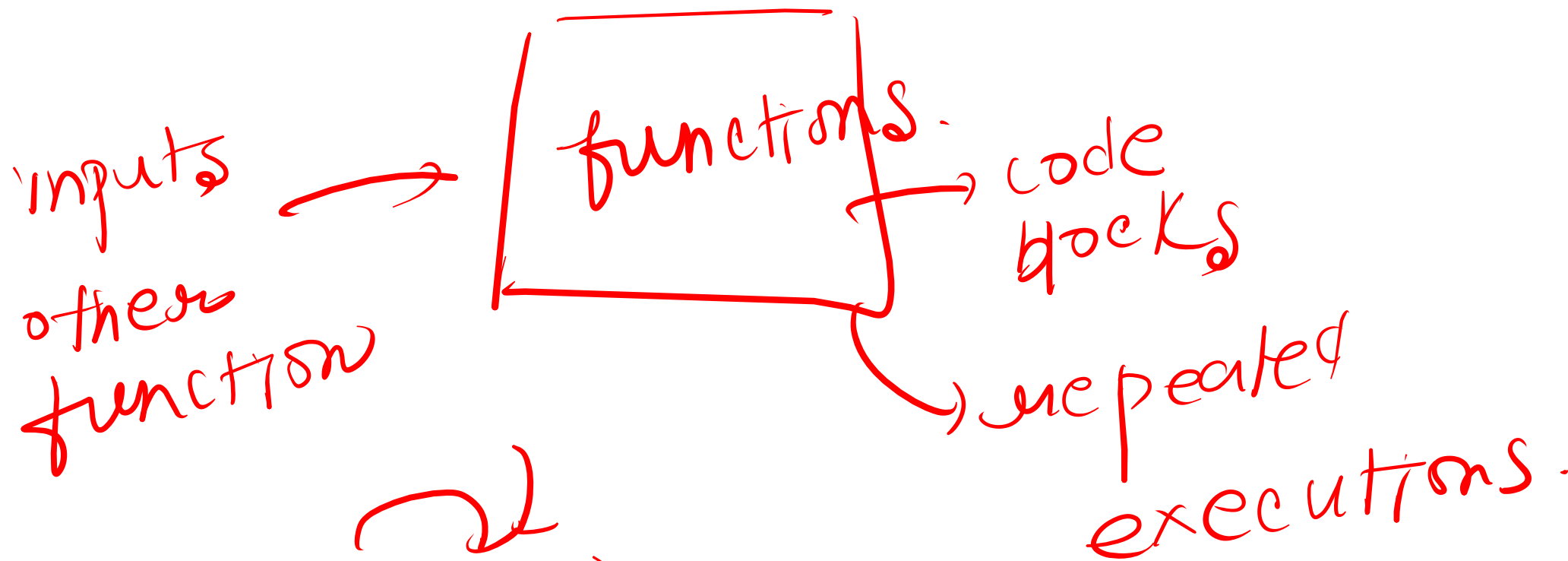


100 test cases.

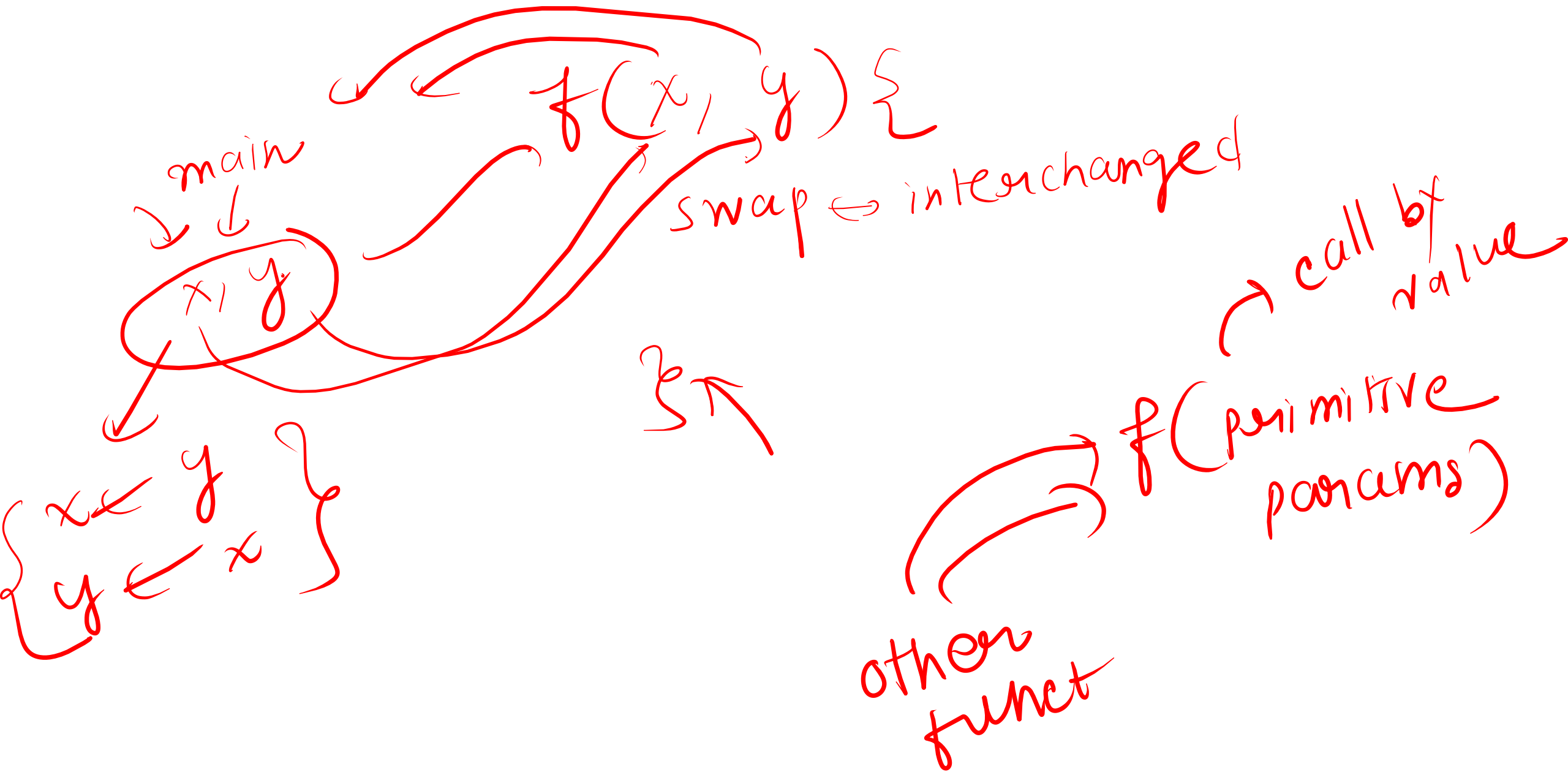
$n=1$
:
10
 n^2



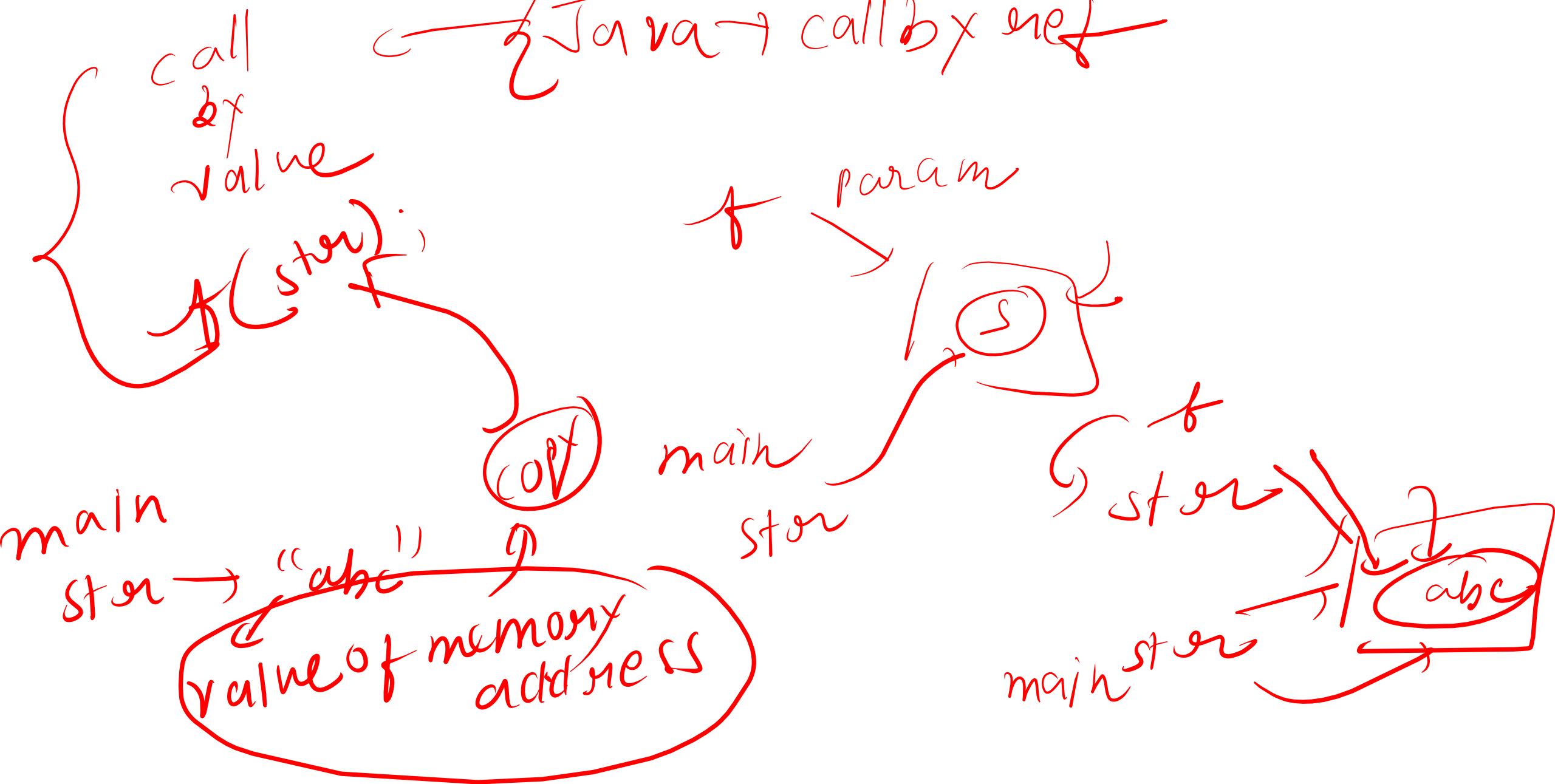




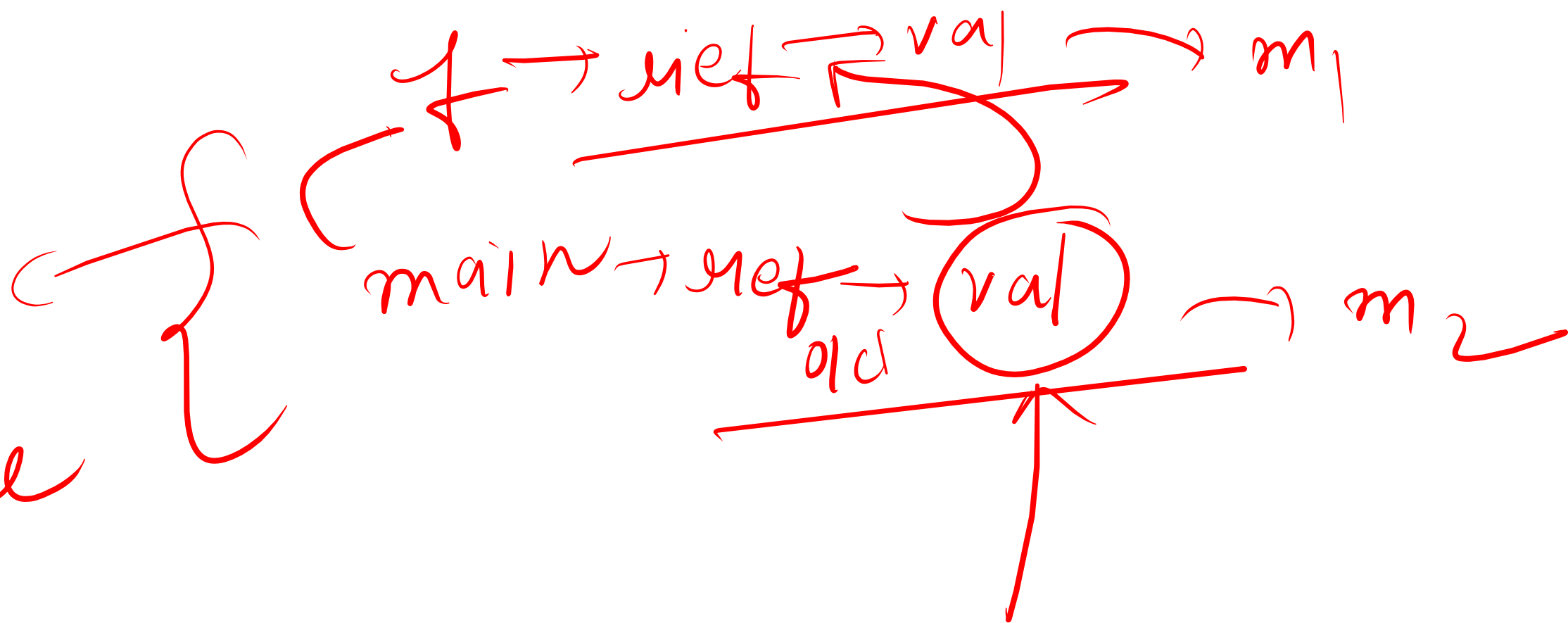
$f(\underline{n})$



Java call by ref

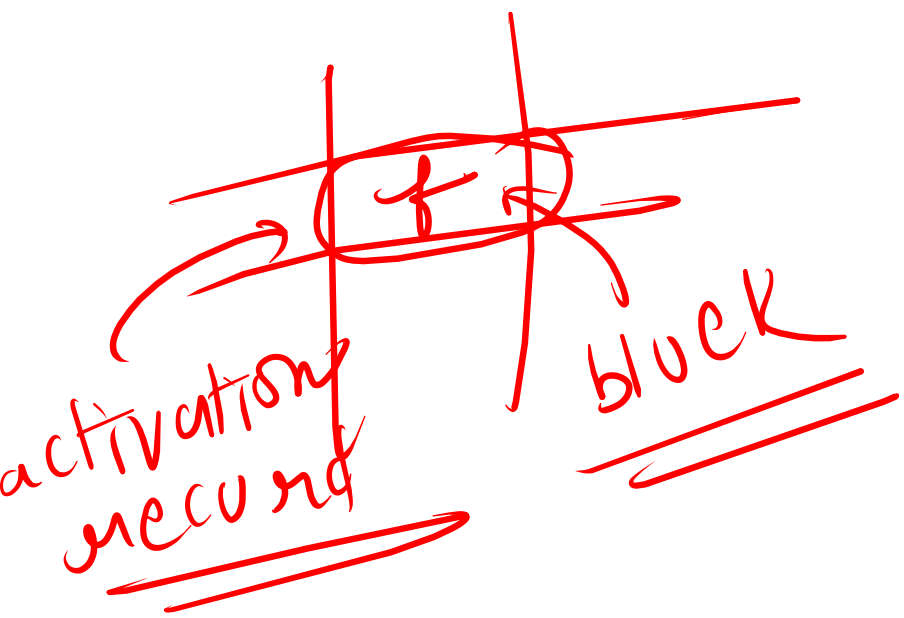


call
of
value



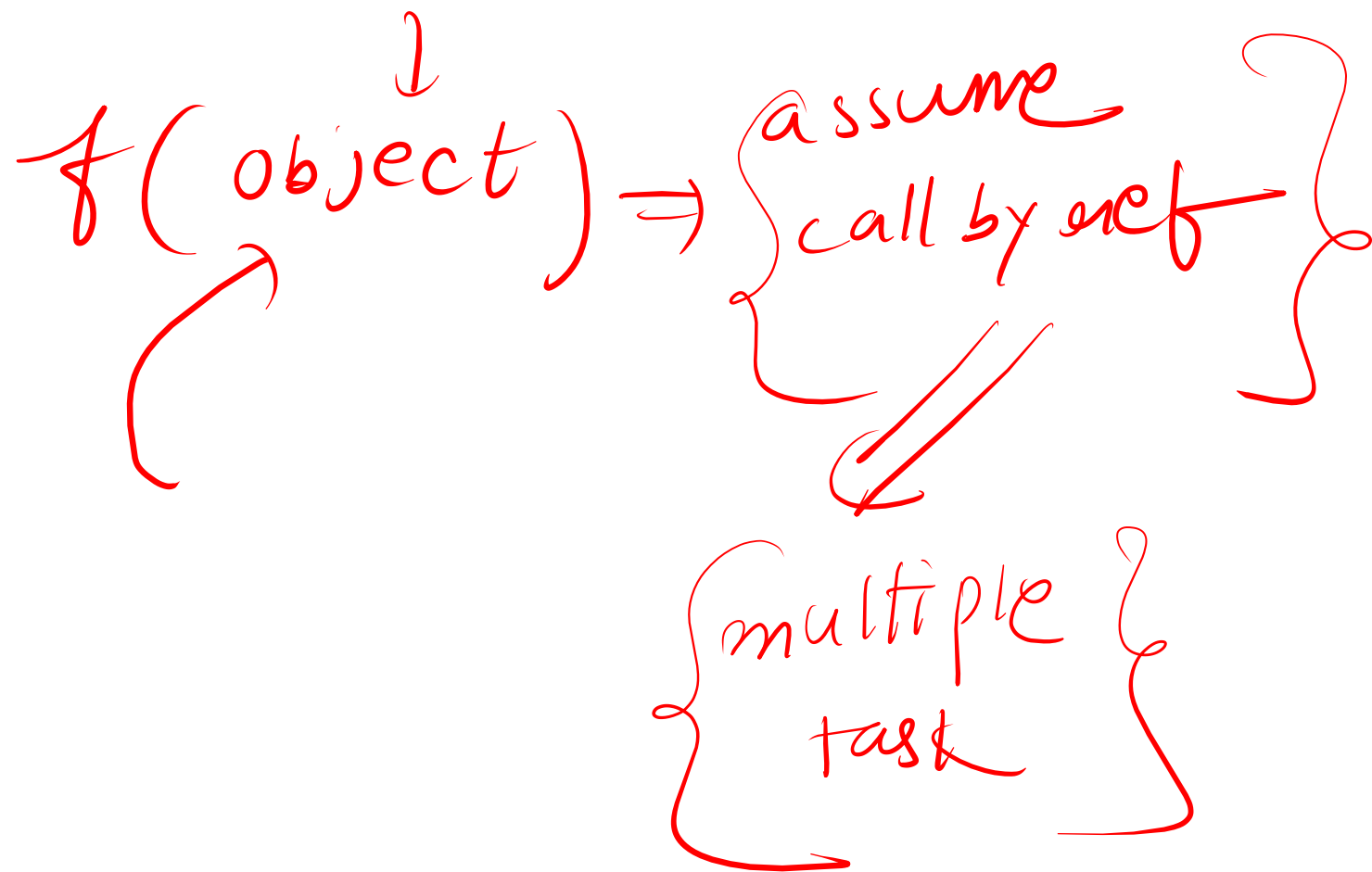
activation record

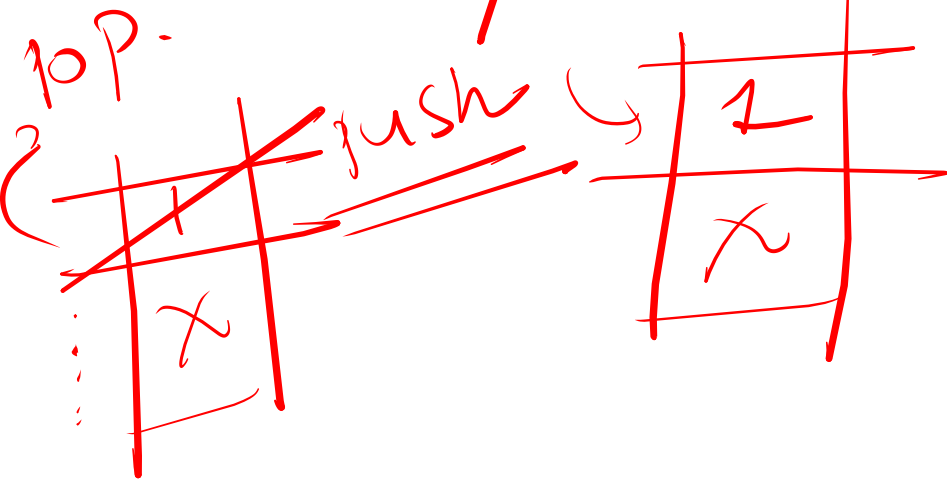
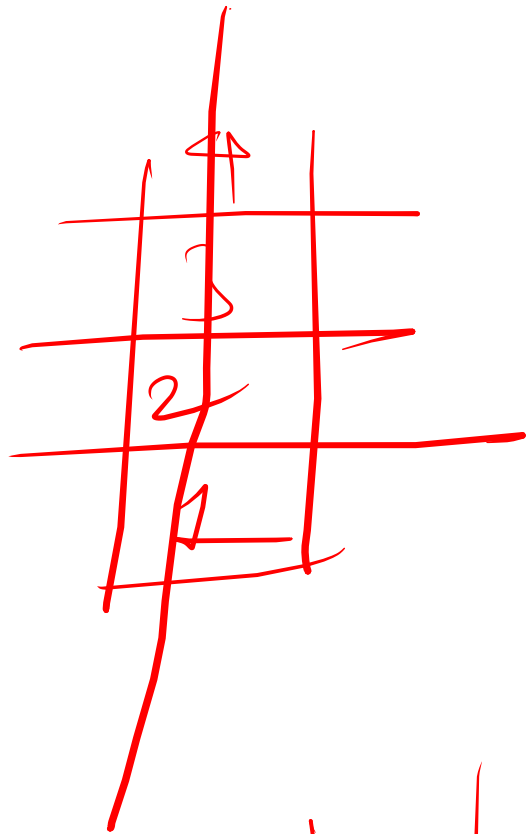
block



$f(\text{object}) \Rightarrow \left\{ \begin{array}{l} \text{assume} \\ \text{call by ref} \end{array} \right\}$

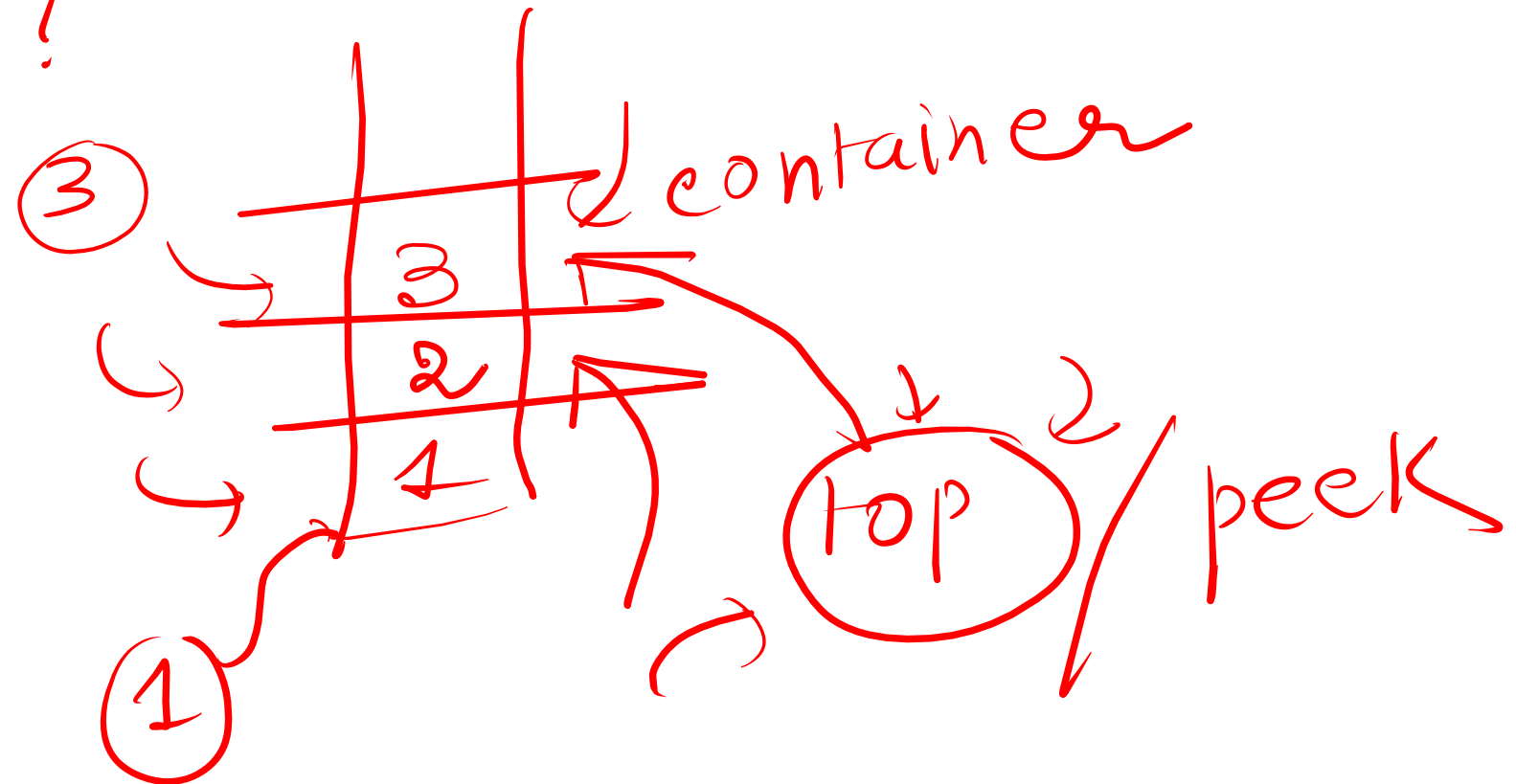
$\left\{ \begin{array}{l} \text{multiple} \\ \text{task} \end{array} \right\}$

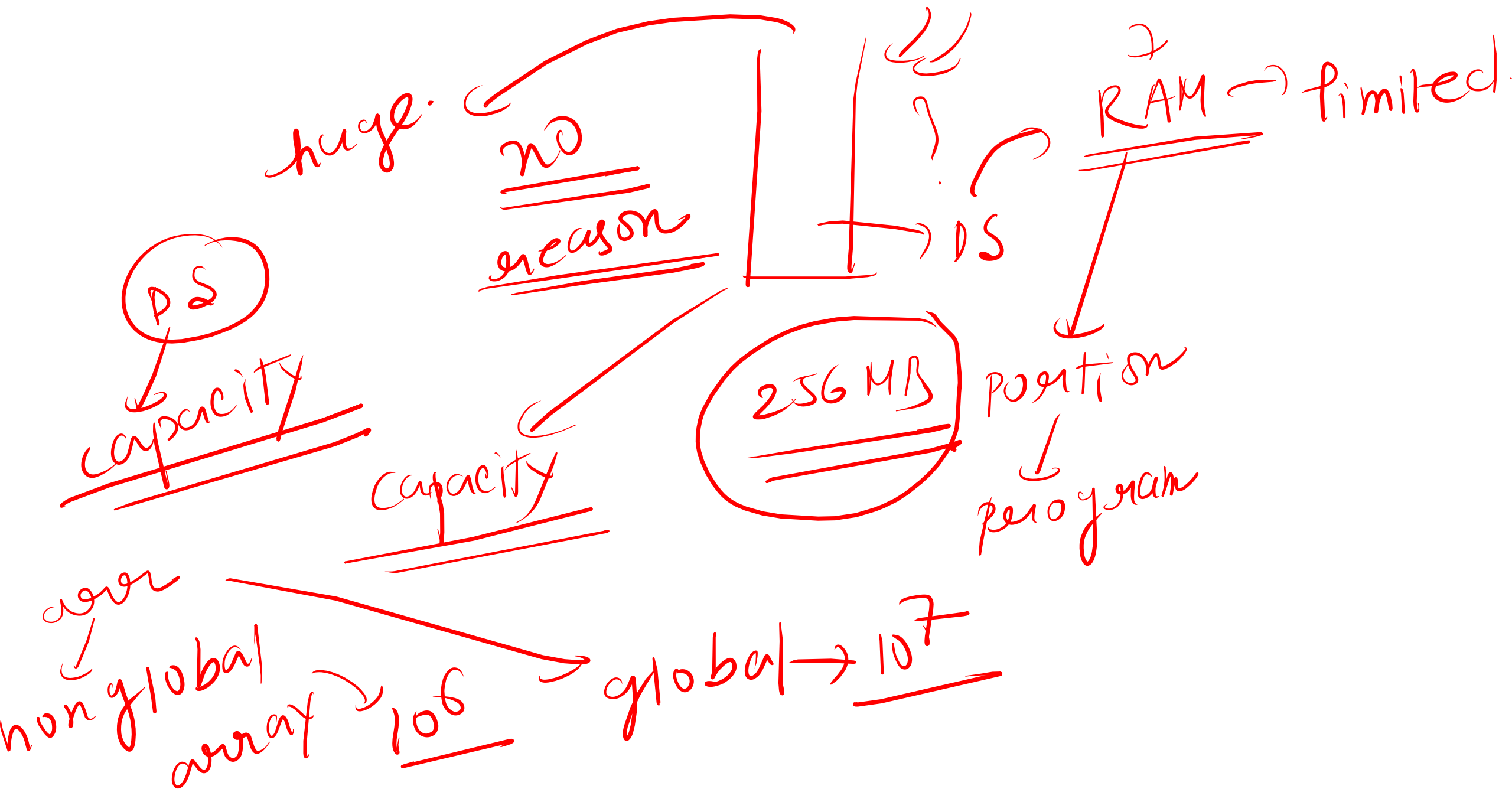


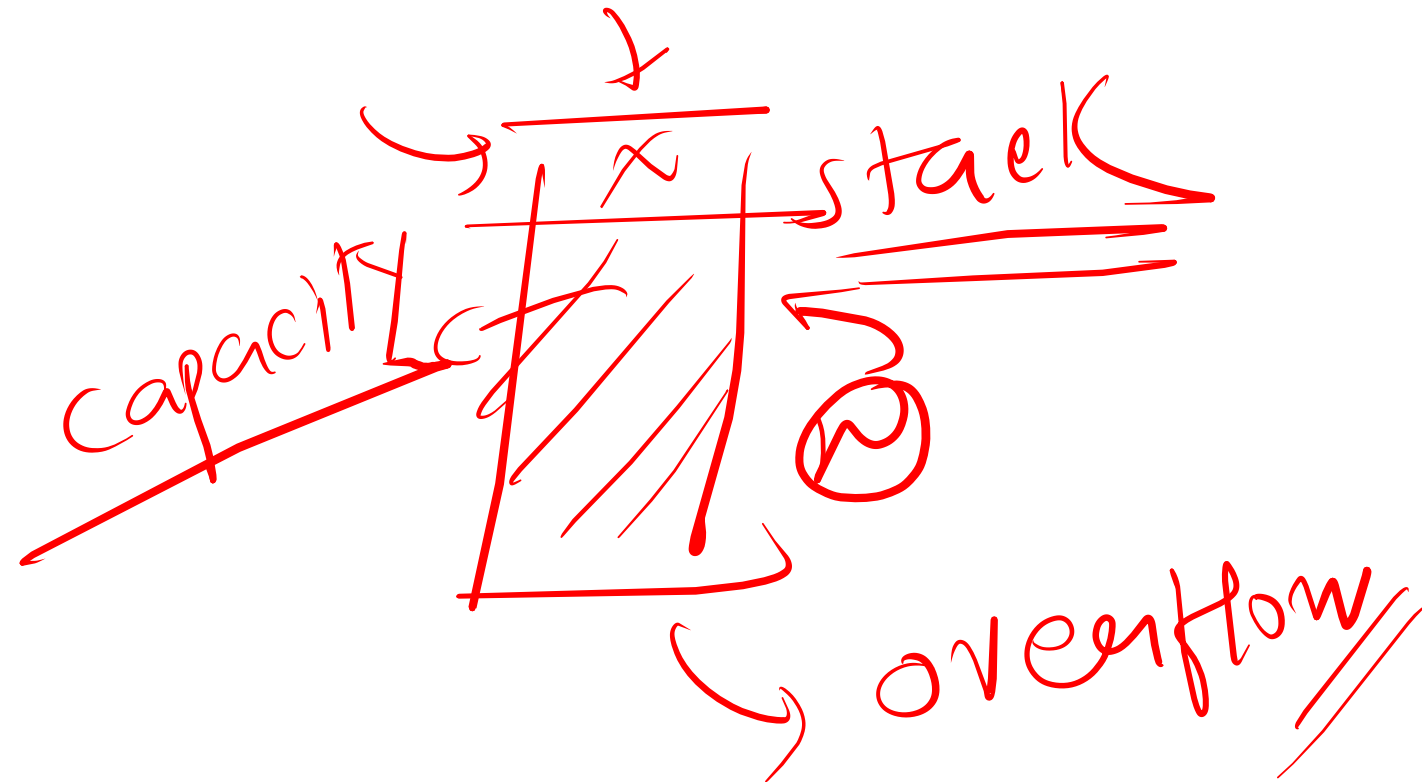


Stack

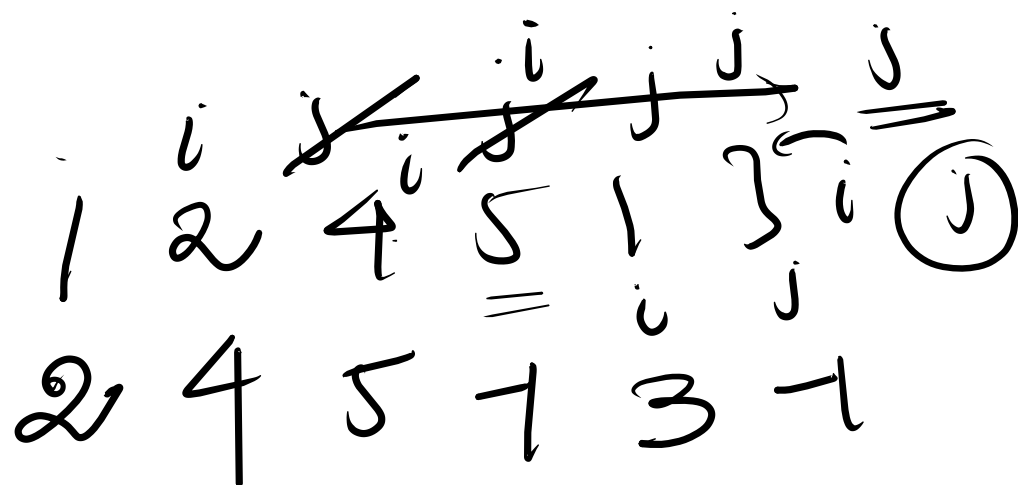
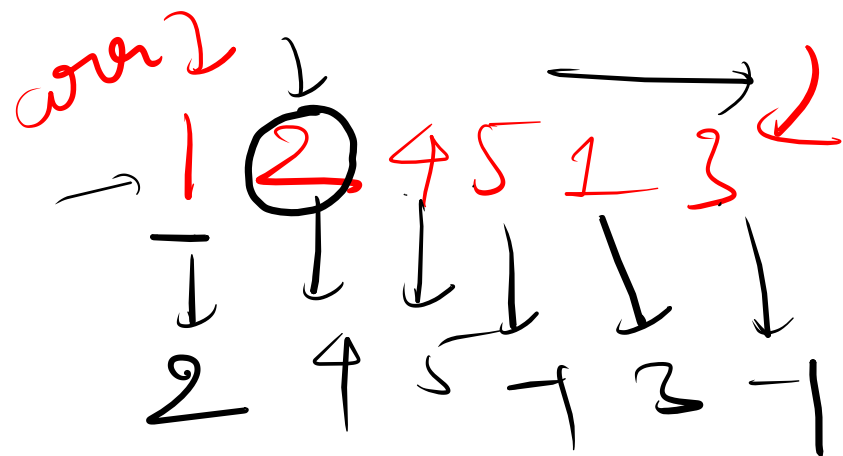
?







next greater element
↓
e



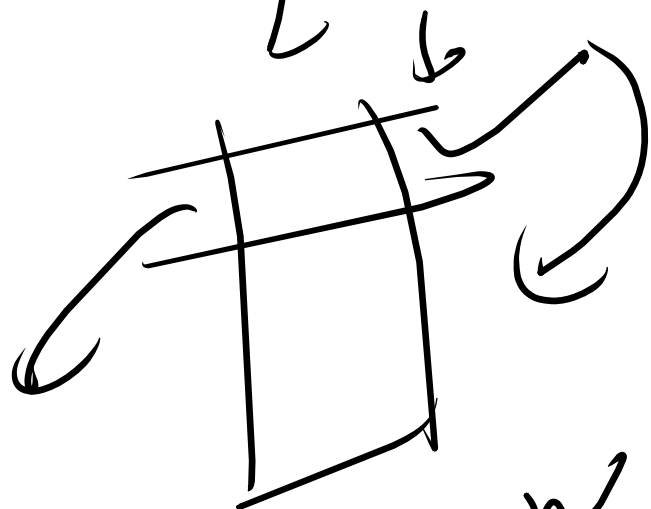
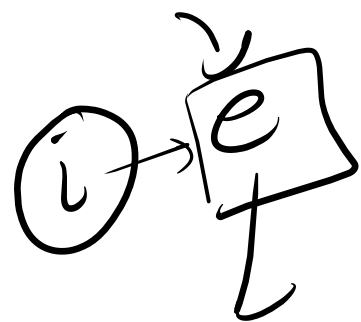
$$\textcircled{i} \quad \underline{j \rightarrow i+1 \dots n-1}$$

$$i \rightarrow \textcircled{n} * \textcircled{n}$$

$$\underline{\underline{O(n^2)}}$$

$$\rightarrow O(1)$$

sc //

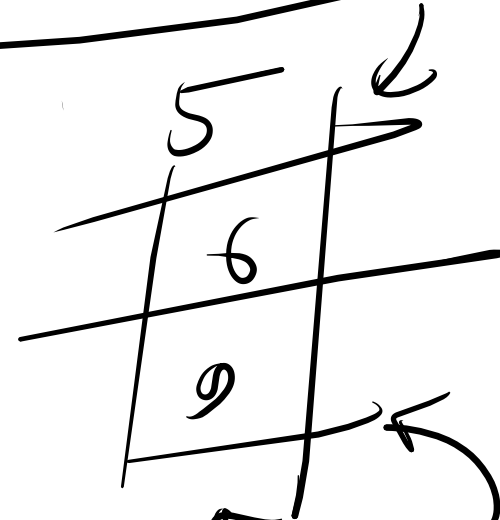
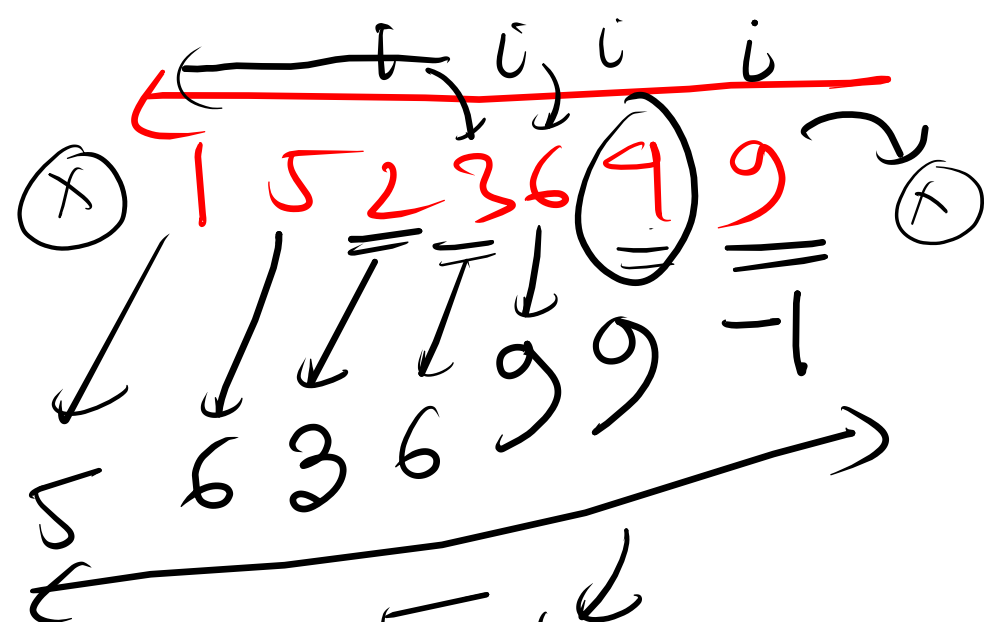


$2n + 2n$

$2n$

$O(\neq n)$

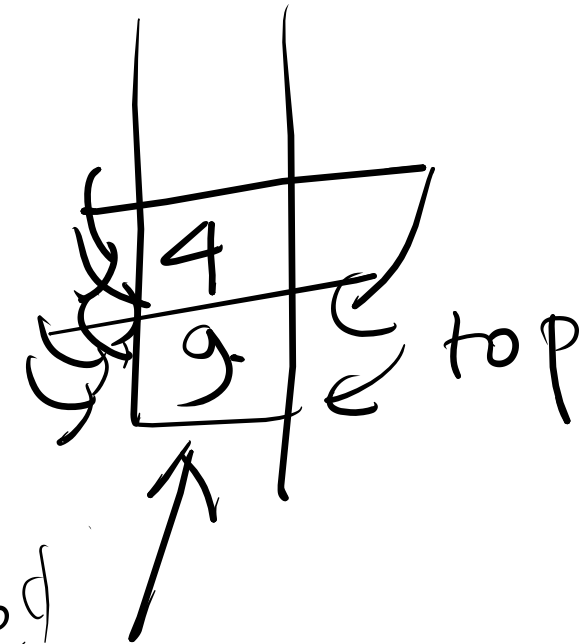
$O(n)$



introduced stack.

n

$O(n) \rightarrow SC$



extra
 $\downarrow$
 $n^2 \rightarrow \textcircled{2}$

extra space } $\textcircled{TC}$

$SC \propto \frac{1}{TC}$ $\downarrow$

7 5 8 2 1 3
 $\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 8 8 $\textcircled{-1}$ 3 3 $\textcircled{-1}$

$\downarrow \downarrow$
 $NG \rightarrow \textcircled{e}$

2
1
3

7
5
8

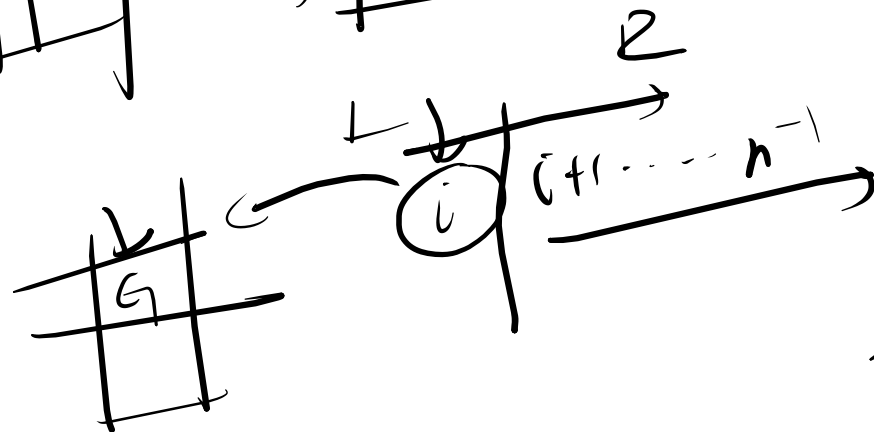
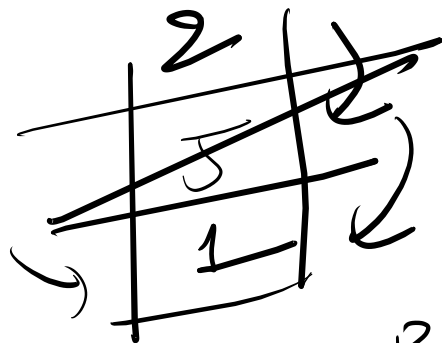
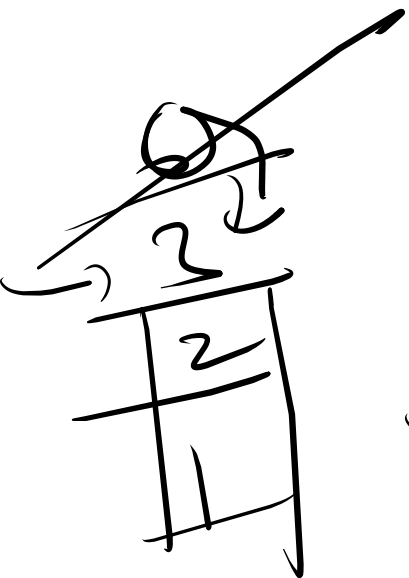
$\textcircled{1}$

```

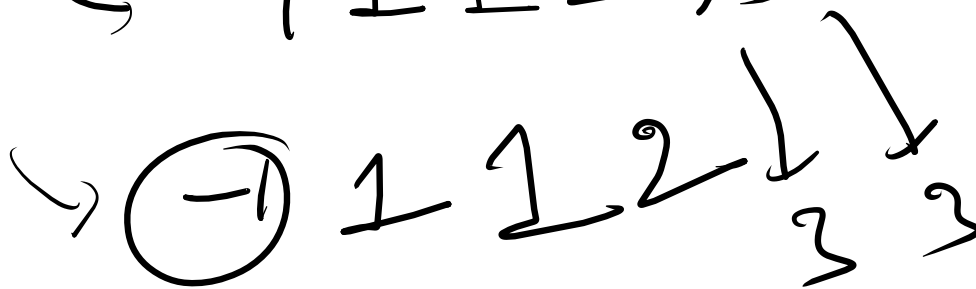
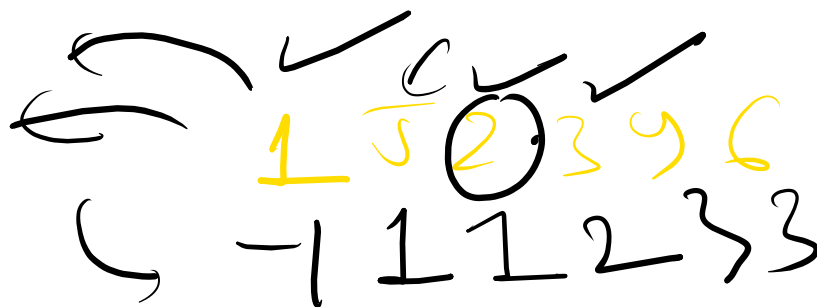
for(i=n-2; i>=0; i--) {
    while(!stk.empty() {
        arr[i] arr[i] >= stk.top();
        stk.pop();
    }
    if(stk.empty()) nge[i] = -1;
    else nge[i] = stk.top();
    stk.push(arr[i]);
}

```

stack <int> stk; arr[n];
 nge[n];
 nge[n-1] = -1;
 stk.push(arr[n-1]);

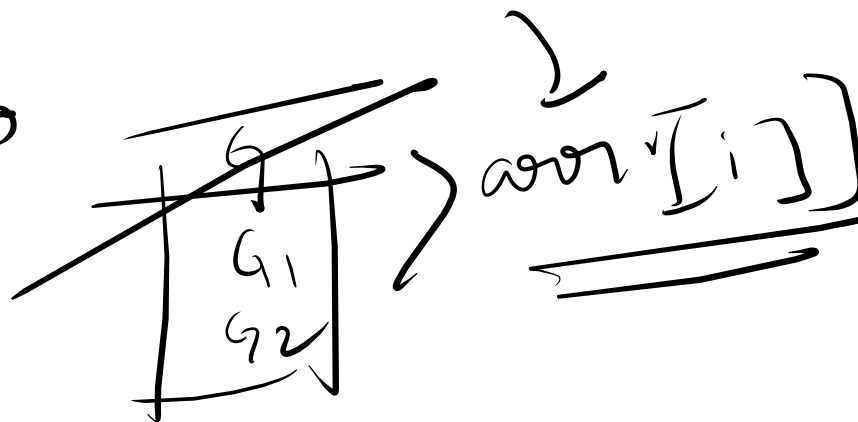


stack



→ PSE

→ e → array

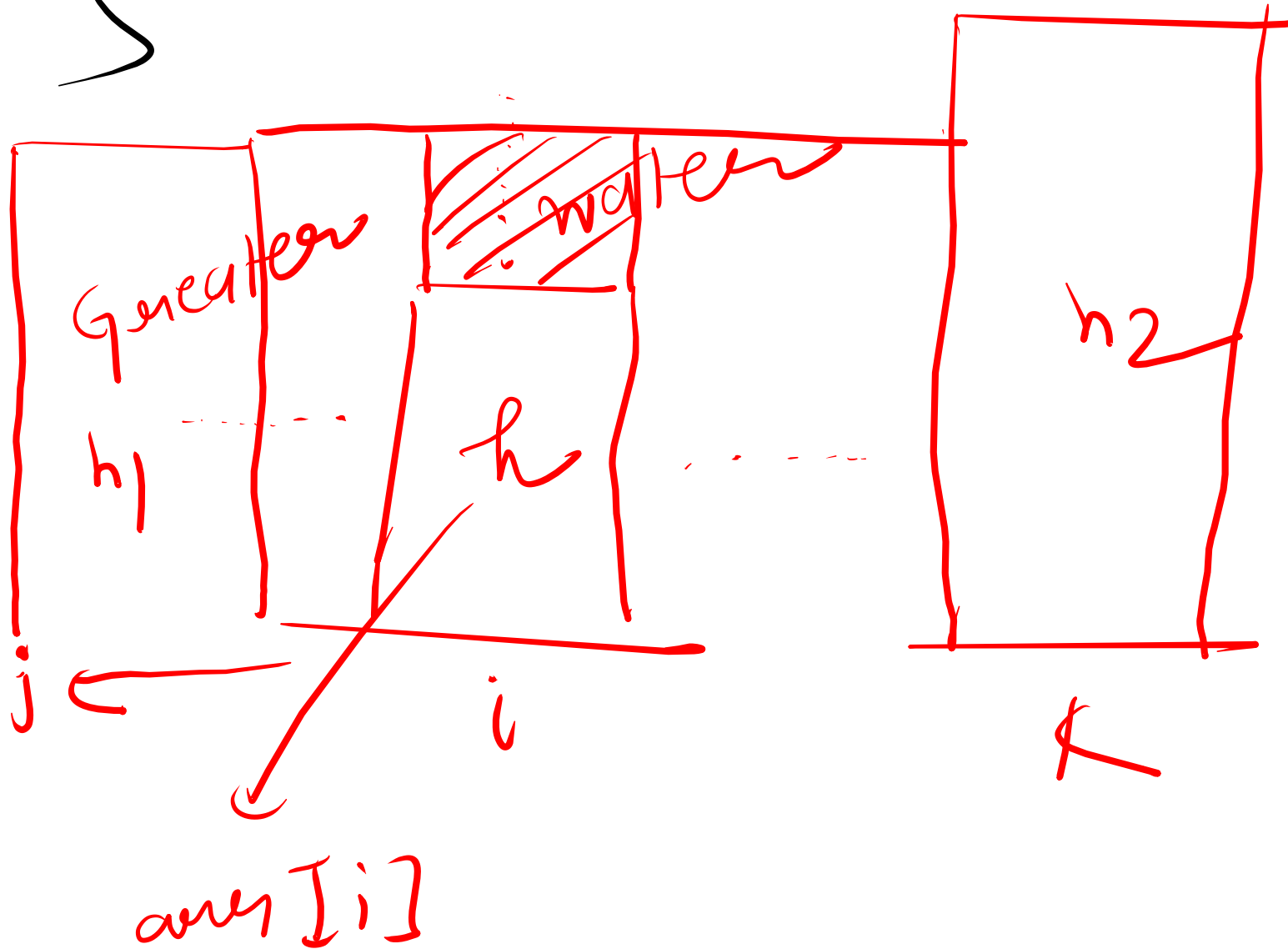


```
arr[n];  
pse[n];  
stack<int> stk;  
pse[0] = -1;  
stk.push(arr[0]);
```

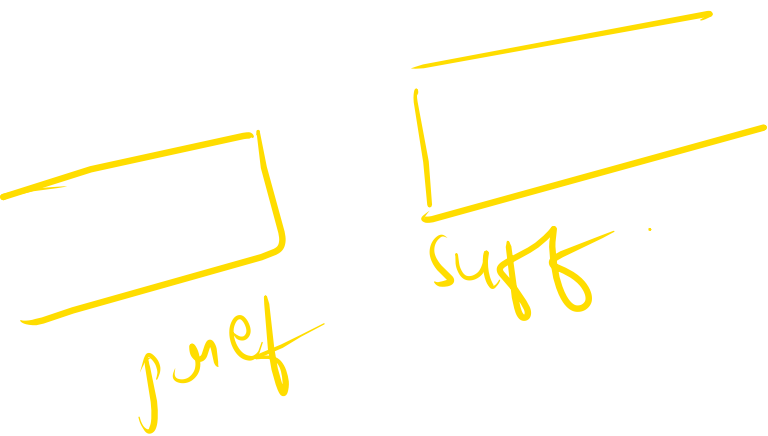
```
for (i = 1; i < n; i++) {  
    while (!stk.empty() &&  
           stk.top() > arr[i])  
        stk.pop();
```

```
    if (stk.empty())  
        pse[i] = -1;  
    else { pse[i] = stk.top(); }  
    stk.push(arr[i]);  
}
```

$$\frac{HSE}{PGE} \rightarrow \underline{\underline{HW}}$$

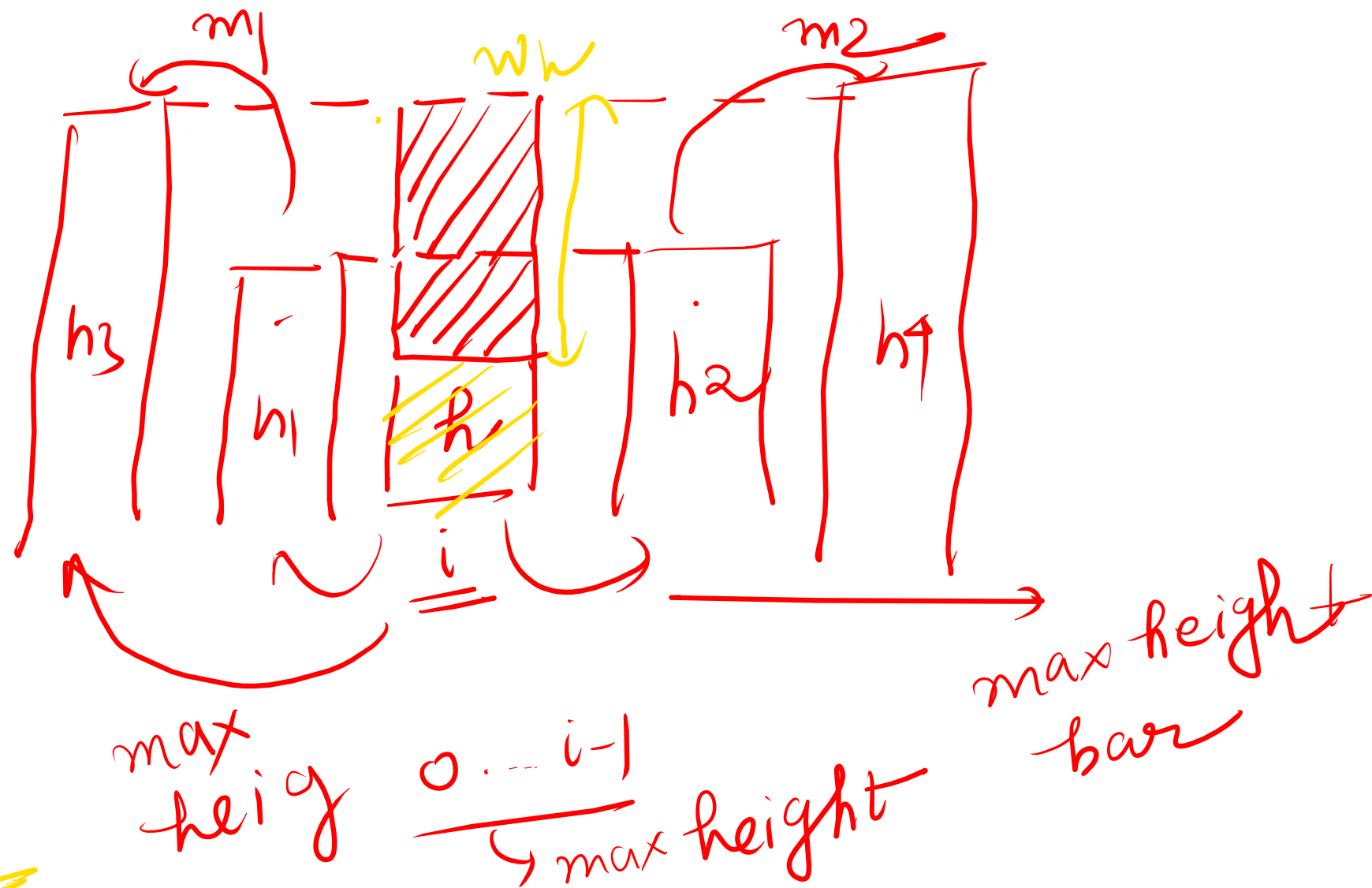


$$\min(m_1, m_2) - h$$



pref $\rightarrow$ pref[i] = 0 $\dots$ i

max

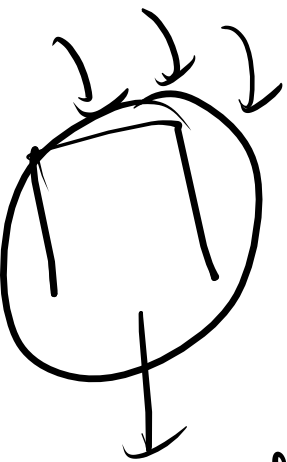


$\text{water}[i]$
 $\rightarrow \text{min}(\text{pref}[i-1], \text{suff}[i+1])$
 $\rightarrow h[i]$

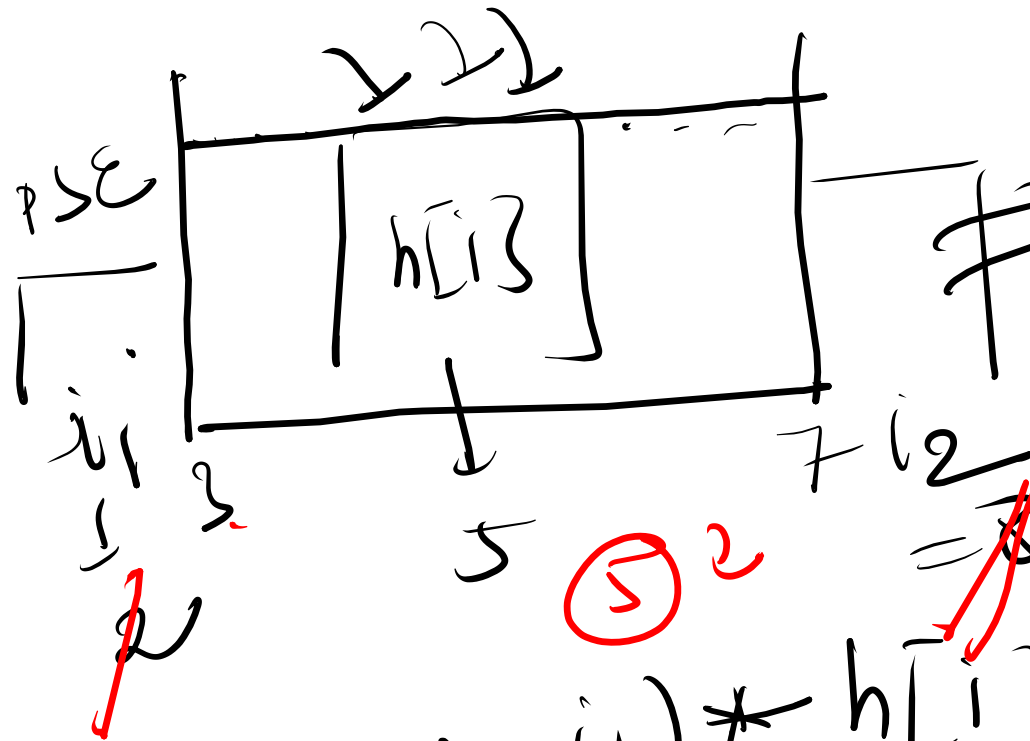
$$\text{suff}[i] = \max(\text{suff}[i+1], h[i]);$$

0 1 2
i n-1
j_max!

$$i \left(\frac{i+1 \dots n-1}{\text{max}} \right)$$



height $\rightarrow$ rectangle

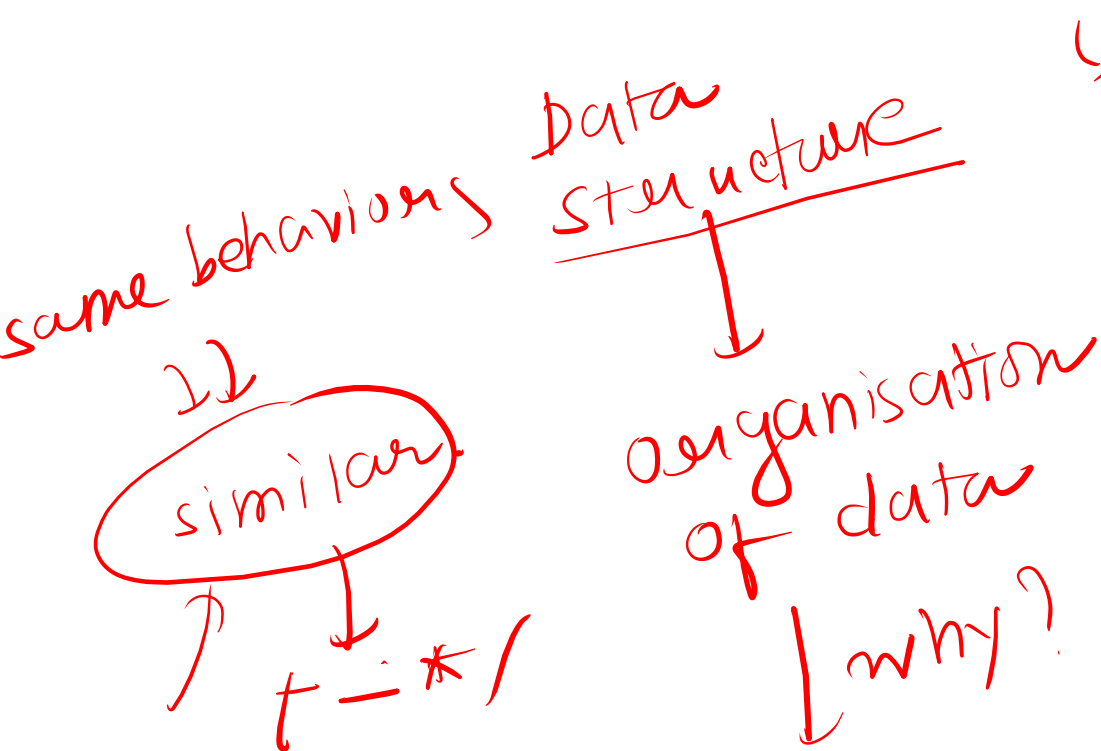


$$(i_2 - i_1 - 1)$$

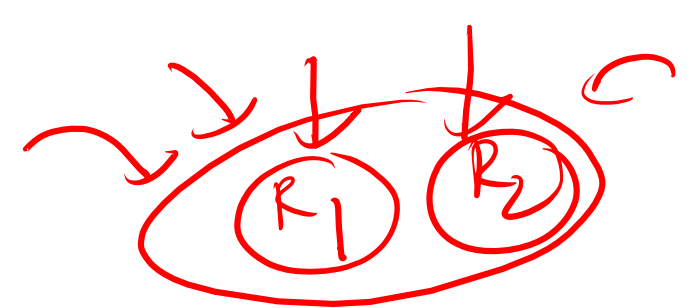
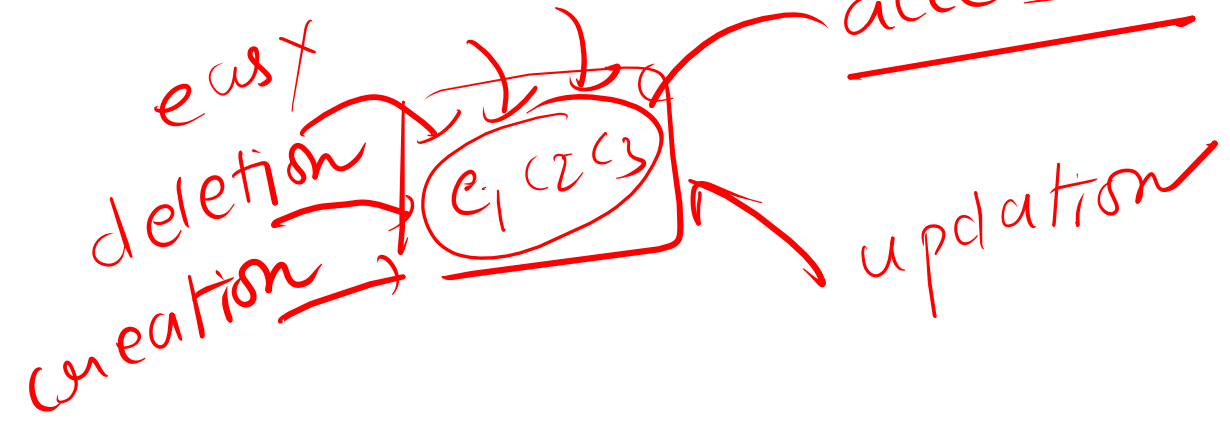
$$(i_2 - i_1) * \frac{h[i]}{2}$$

area $\rightarrow$ ans

$$(i_2 - i_1) * h[i] \rightarrow \text{area} \rightarrow \text{ans}$$



Array

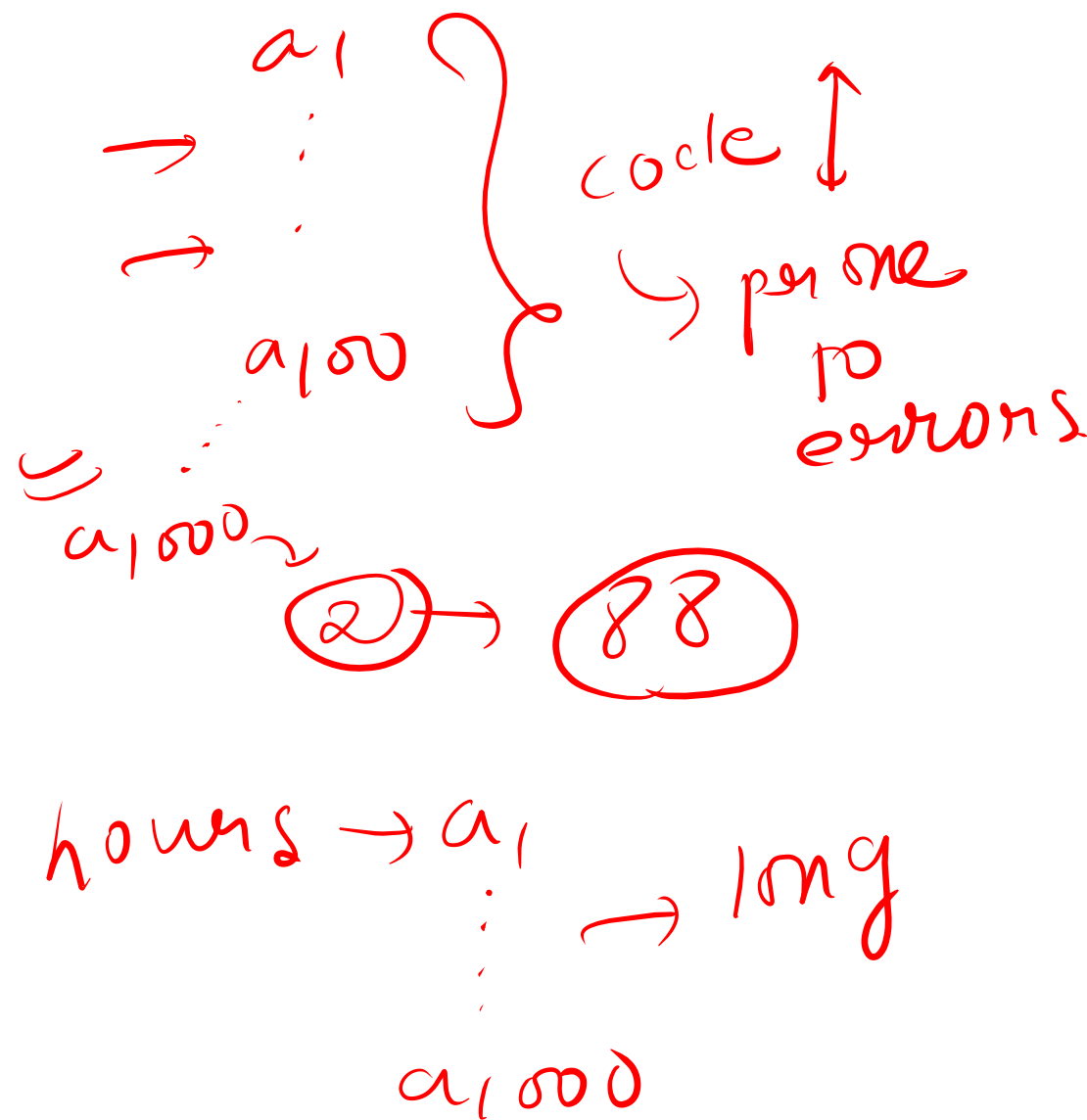
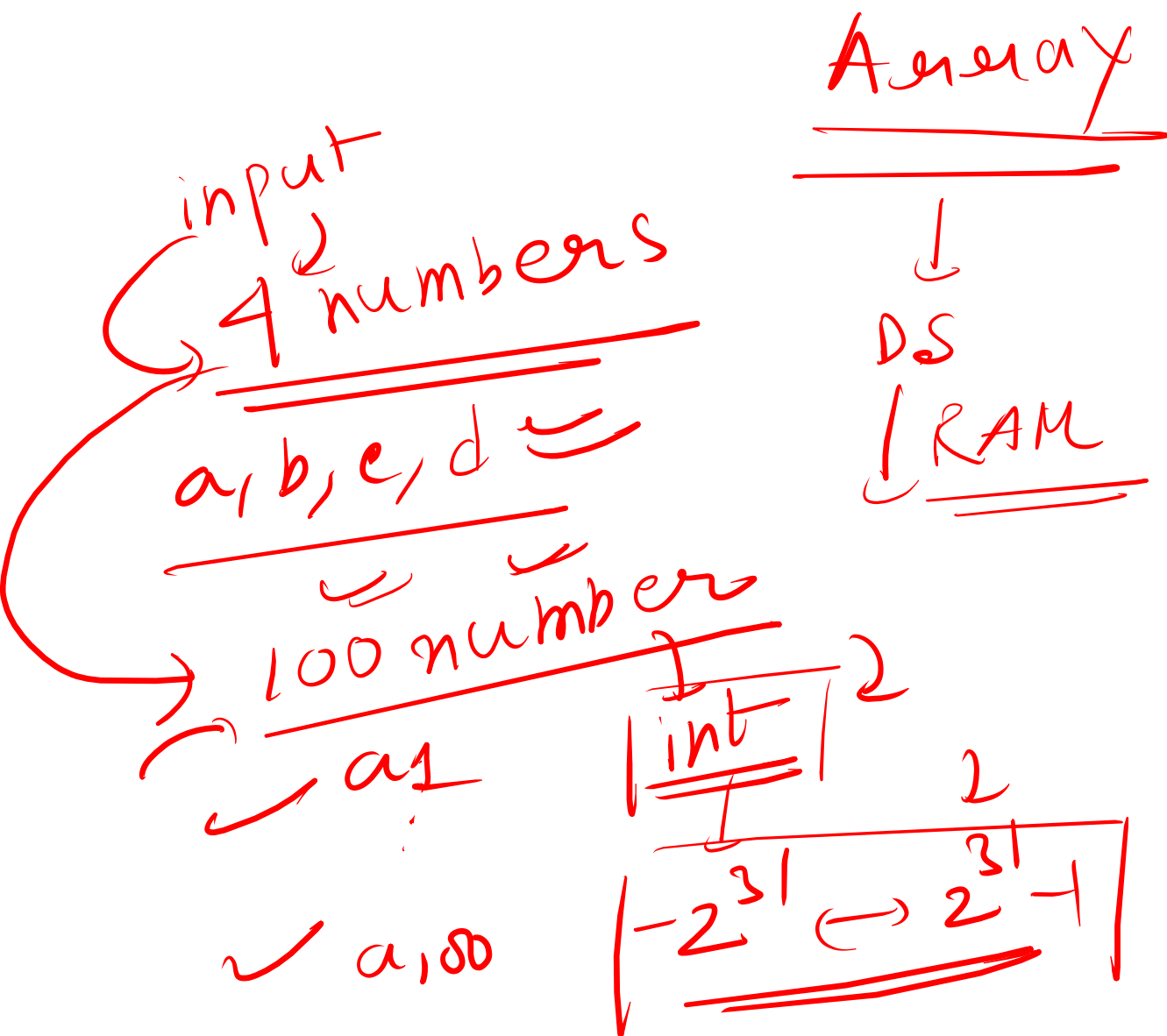


efficient

CRUD

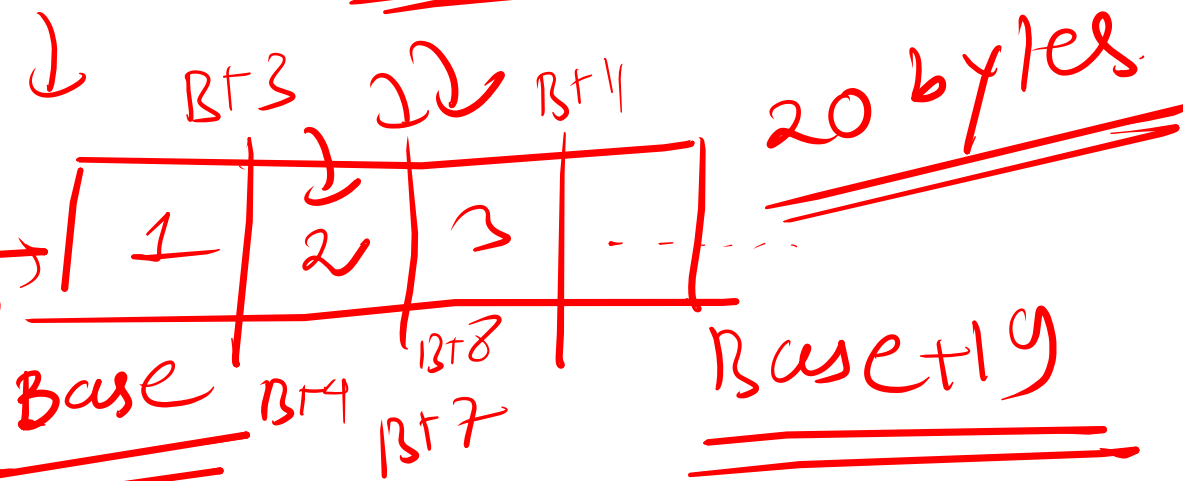
Data structure





`int arr[5];`

`arr` → 5 no. → 5 x 4 bytes?
int ↓



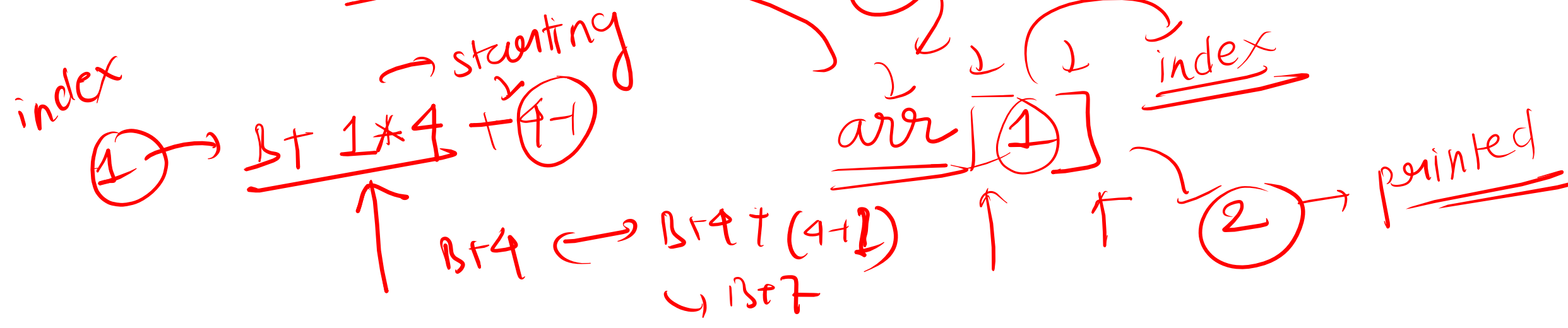
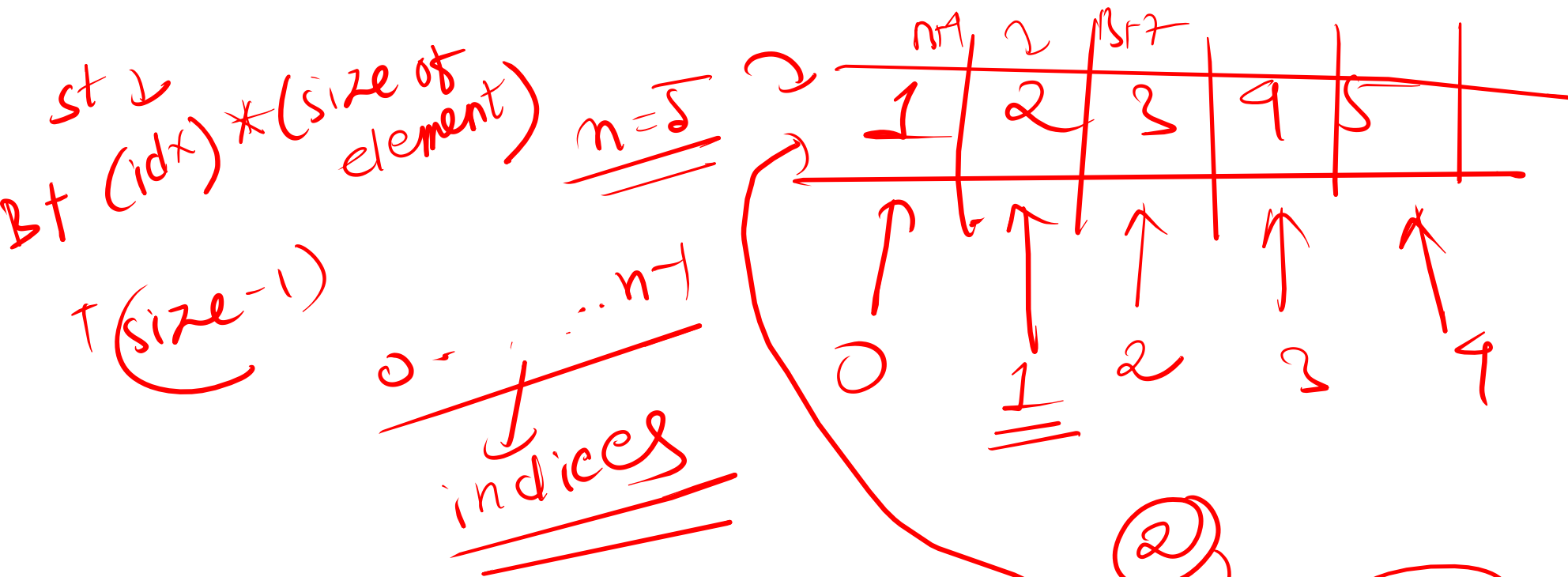
Base? → hexadecimal

② → B+4 ... B+7?

arr

`num` → 4 bytes.

B → B+3 → 1
B+4 ... B+7 → 2 ...

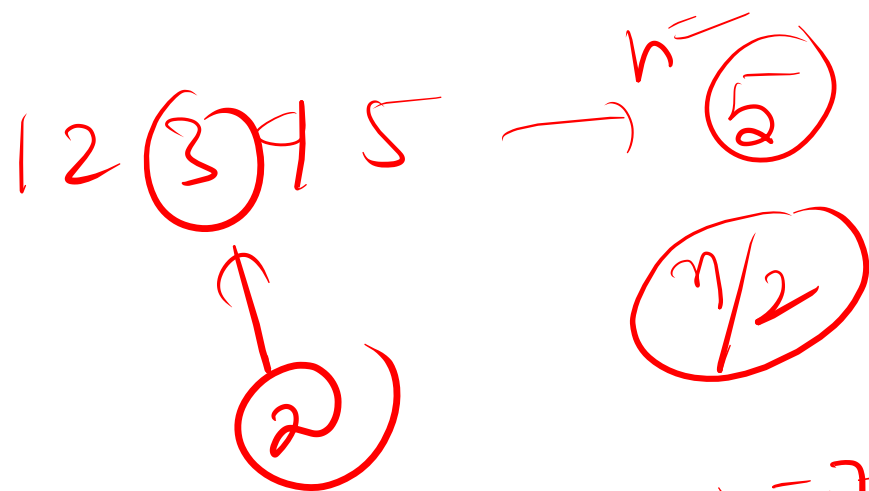
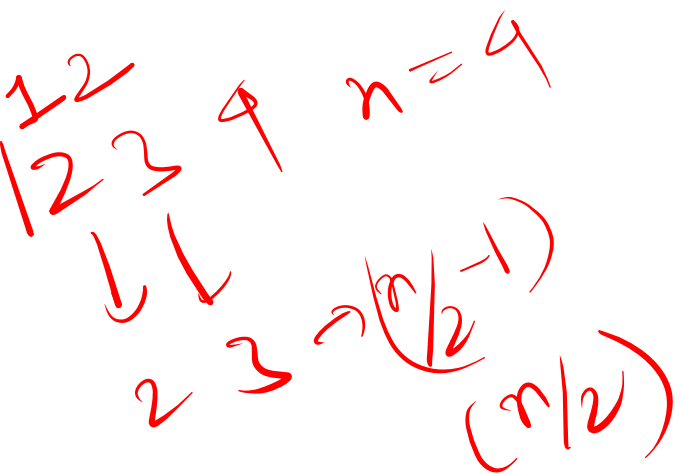


arr → 1 2 (3) 4 5 ↙ console
input.

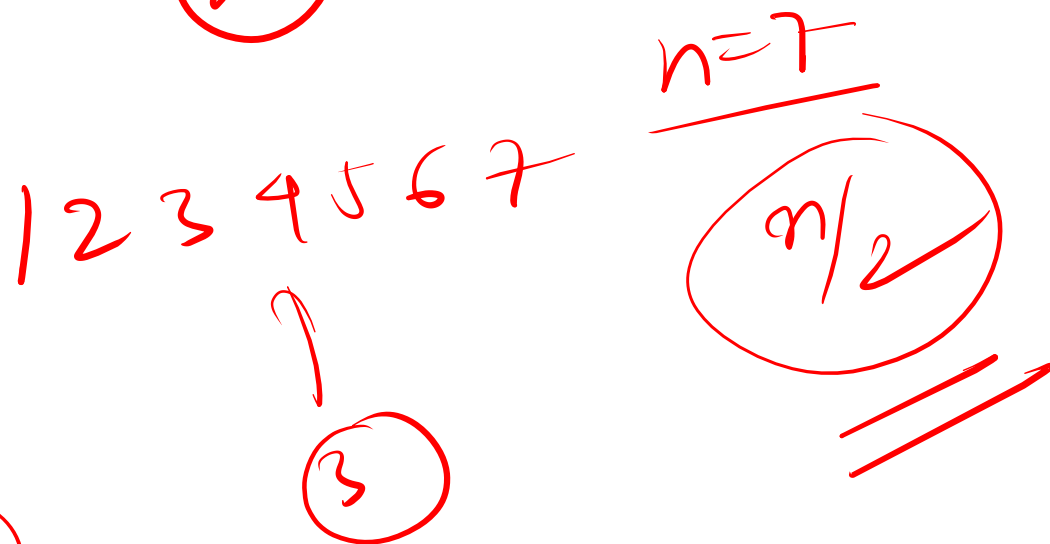
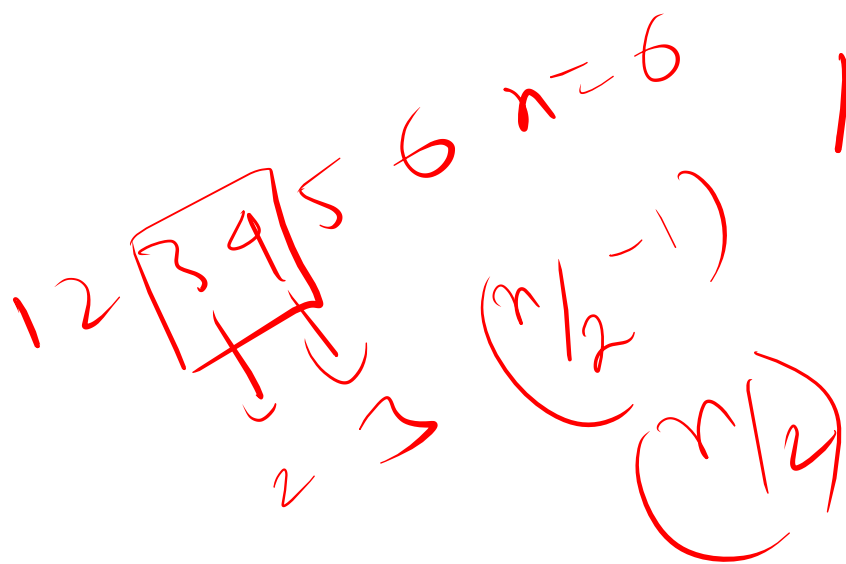
{ mid
element(s)

↓
(3) ✓

1 2 (3 4) 5 6
↓
3 4



$\left. \begin{matrix} \rightarrow \text{integer} \\ \text{div. ?} \end{matrix} \right\}$



$0 \rightarrow n-1$
 $1 \rightarrow n-2$
 $2 \rightarrow n-3$
 $3 \rightarrow n-4$

mid-1 / mid

$1\ 2\ 3\ 4\ 5 \Rightarrow \underline{\underline{5\ 4\ 3\ 2\ 1}}$

$\begin{array}{ccccccccc} & & j & i & j & i & & & \\ & & \downarrow & \downarrow & \downarrow & \downarrow & & & \\ 1 & 2 & 3 & 4 & 5 & & & & \\ \uparrow & \uparrow & \downarrow & \downarrow & \downarrow & & & & \\ i & i & j & j & j & & & & \end{array} \Rightarrow 5\ 4\ 3\ 2\ 1$

$\begin{array}{ccccc} & 5 & 4 & 3 & 2 & 1 \\ \uparrow & \leftarrow & & & & \rightarrow \end{array}$

$\{ \underline{i / j} \}$
 $\{ \underline{i = j} \}$

$\begin{array}{ccc} & i & j \\ \leftarrow & & \rightarrow \\ 1 & 2 & 3 \end{array}$

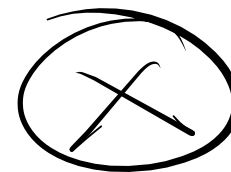
i

$\begin{array}{ccc} & j & \\ \leftarrow & & \rightarrow \\ 4 & 5 & 6 \end{array}$

j

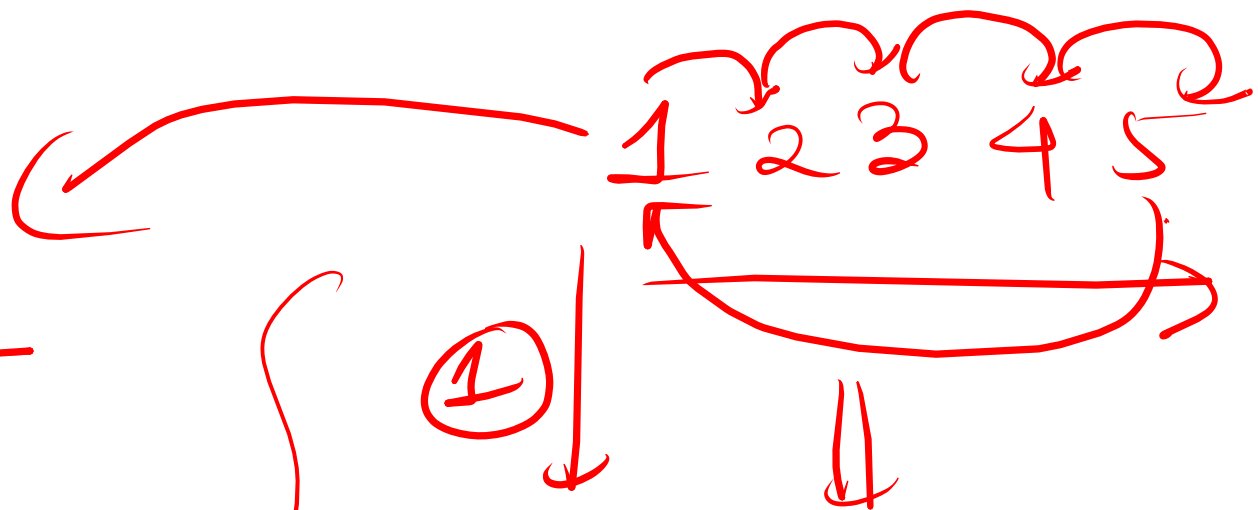
$\begin{array}{cccccc} & j & & i & & \\ \leftarrow & & & & & \rightarrow \\ i & i & i & j & j & j \\ \underline{1} & 2 & 3 & 4 & 5 & \underline{6} \end{array}$

$\begin{array}{ccc|ccc} \leftarrow & & & & & \rightarrow \\ & j & & i & & \\ 6 & 5 & 4 & 3 & 2 & 1 \\ \leftarrow & & & & & \rightarrow \end{array}$



$i > j$

5 1 2 3 4



$n=5$

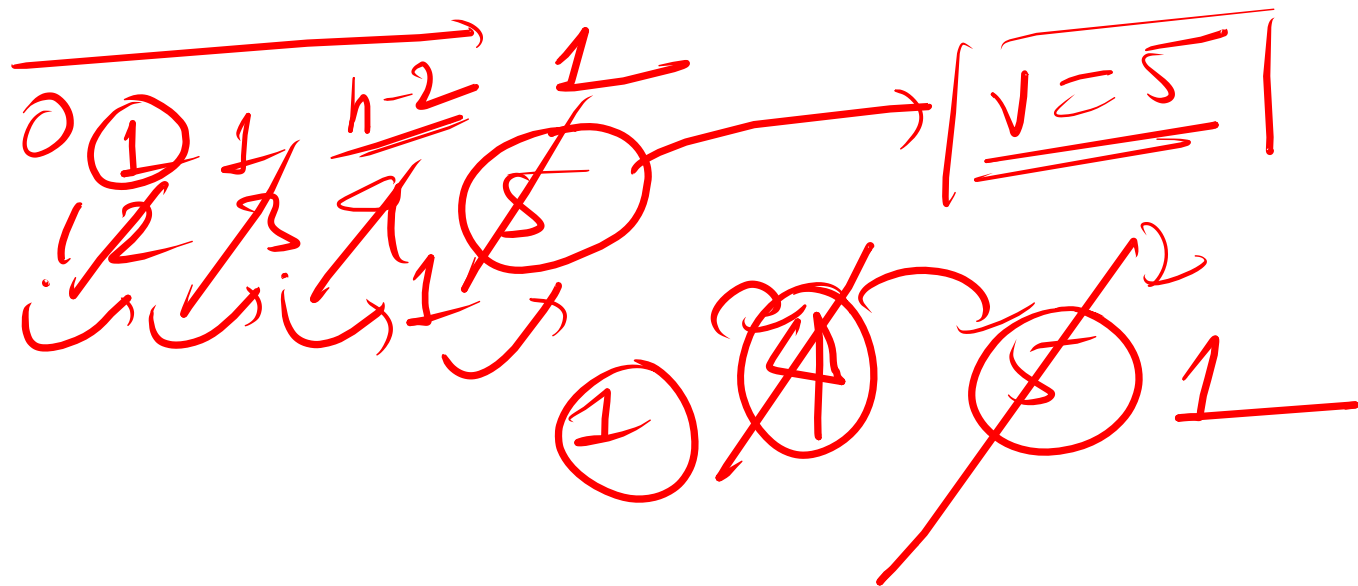
clockwise
direction
by one
step

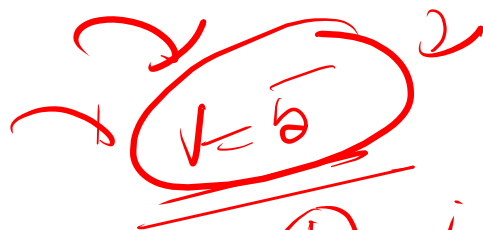
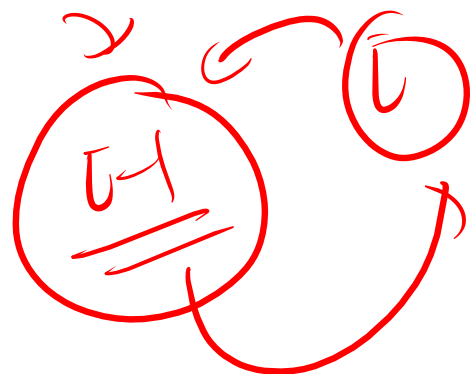
5 1 2 3 4

4
↓
lost

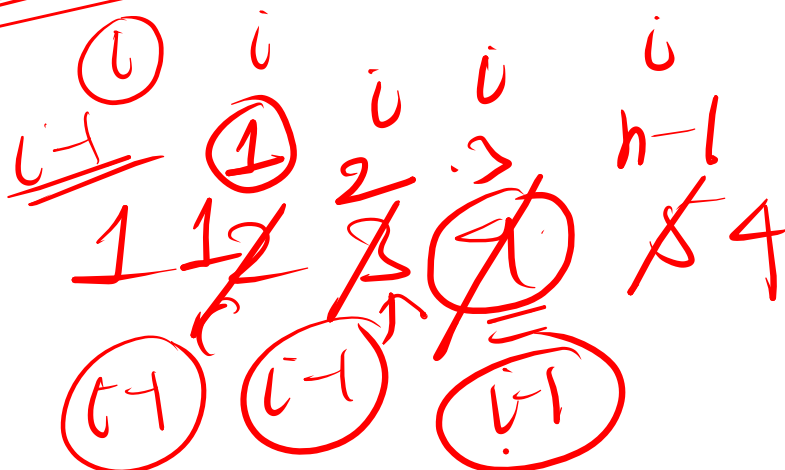
3
↓
lost

2
↓
lost

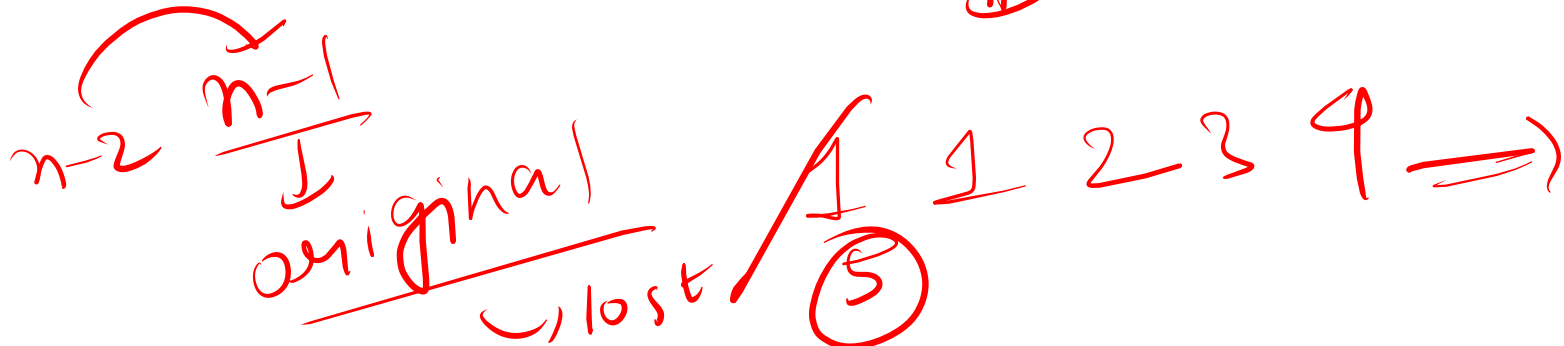
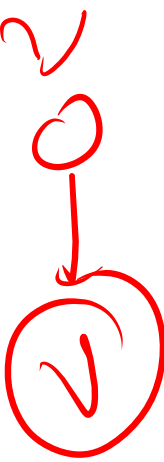
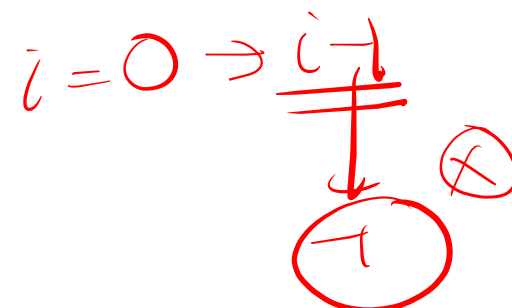




i



$i=0$ (X)



5 1 2 3 4

i $i+1$ $i-1$ i

for ($i = n-1$; $i \geq 1$; $i--$)
 $arr[i] = arr[i-1];$

$i-1$

$v = arr[n-1]$

Rotated
arr.

$n-1$
0

1 2 3 ... $n-1$

$\otimes$

$arr[0] = v;$